Introduction to Electrical Circuits Lecture-6

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16th March 2025

Reference Books

Recommended

- R. L. Boylestad, “Introductory Circuit Analysis”, 11th Edition, Pearson Education, Inc.
- B. L. Theraja, A. K. Theraja, “A Textbook of ELECTRICAL TECHNOLOGY in SI Units Volume II, AC & DC Machines”, S. Chand & Company Ltd., (Multicolor illustrative Edition).

Supplementary

- Charles K. Alexander & Mathew N.O. Sadiku, “Fundamentals of Electric Circuits”, 3rd edition, The McGraw-Hill companies.

Chapter 8

Nodal Analysis

1. Determine the number of nodes within the network.

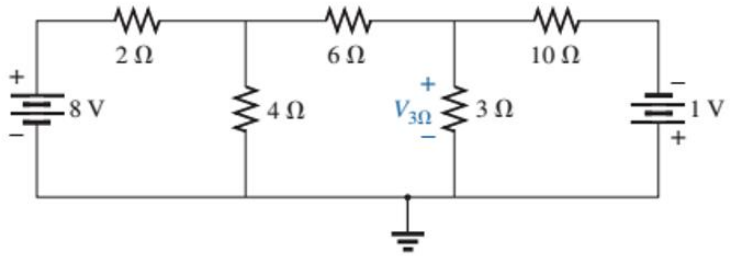
2. Pick a reference node, and label each remaining node with a subscripted value of voltage: V1, V2, and so on.

3. Apply Kirchhoff's current law at each node except the reference. Assume that all unknown currents leave the node for each application of Kirchhoff's current law. In other words, for each node, don't be influenced by the direction that an unknown current for another node may have had. Each node is to be treated as a separate entity, independent of the application of Kirchhoff's current law to the other nodes.

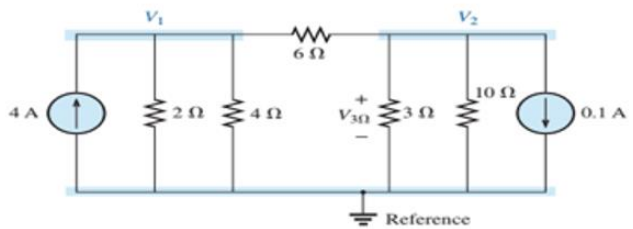
4. Solve the resulting equations for the nodal voltages.

8.9 NODAL ANALYSIS

EXAMPLE 8.24 Find the voltage across the 3 Ω resistor in Fig. 8.61 by nodal analysis.



Solution: Converting sources and choosing nodes (Fig. 8.62), we have



$$\left(\frac{1}{2\,\Omega}+\frac{1}{4\,\Omega}+\frac{1}{6\,\Omega}\right)V_1-\left(\frac{1}{6\,\Omega}\right)V_2=+4\text{ A}$$

$$\left(\frac{1}{10\,\Omega}+\frac{1}{3\,\Omega}+\frac{1}{6\,\Omega}\right)V_2-\left(\frac{1}{6\,\Omega}\right)V_1=-0.1\text{ A}$$

$$\frac{11}{12}V_1-\frac{1}{6}V_2=4$$

$$-\frac{1}{6}V_1+\frac{3}{5}V_2=-0.1$$

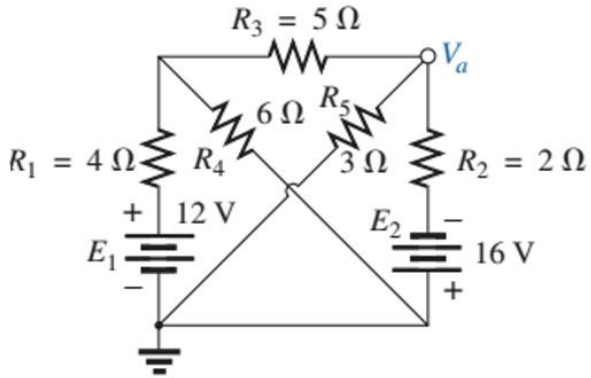
$$11V_1-2V_2=+48$$

$$-5V_1+18V_2=-3$$

$$V_2=V_{3\Omega}=\frac{\begin{vmatrix}11 & 48\\ -5 & -3\end{vmatrix}}{\begin{vmatrix}11 & -2\\ -5 & 18\end{vmatrix}}=\frac{-33+240}{198-10}=\frac{207}{188}=\mathbf{1.10\text{ V}}$$

Exercise Problems

32. Using mesh analysis, determine $I_{5\Omega}$ and V_a for the network in Fig. 8.121(b).



(b)

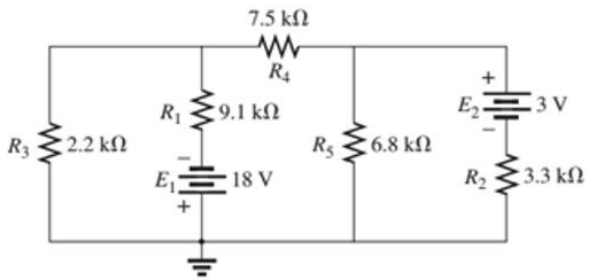
Solution:

From Sol. 24(b)
I_1 I_2 I_3

$I_1(6 + 4) - 4I_2 = -12$
$I_2(4 + 5 + 2) - 4I_1 - 2I_3 = 12 + 16$
$I_3(2 + 3) - 2I_2 = -16$

$I_{5\Omega} = I_2 = \mathbf{1.95\ A}$
$I_3 = -2.42\ \text{A},\ \therefore\ V_a = (I_3)(3\ \Omega) = (-2.42\ \text{A})(3\ \Omega) = \mathbf{-7.26\ V}$

33. Using mesh analysis, determine the mesh currents for the networks in Fig. 8.122.



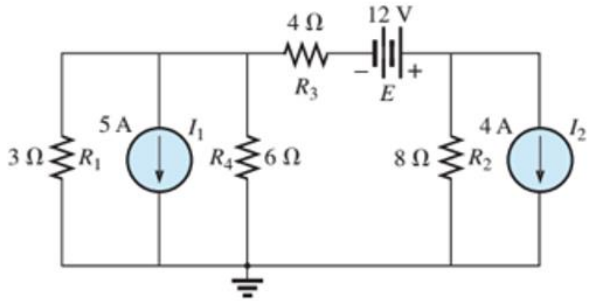
(I)

(I): I_1 I_2 I_3

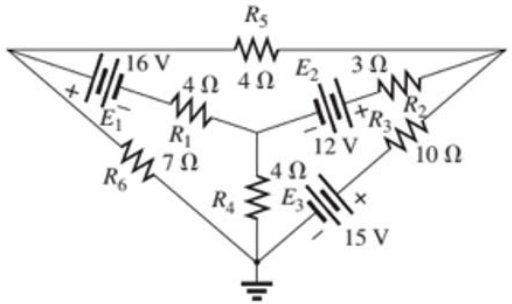
$(2.2\ \text{k}\Omega + 9.1\ \text{k}\Omega)I_1 - 9.1\ \text{k}\Omega I_2 = 18$
$(9.1\ \text{k}\Omega + 7.5\ \text{k}\Omega + 6.8\ \text{k}\Omega)I_2 - 9.1\ \text{k}\Omega\ I_1 - 6.8\ \text{k}\Omega I_3 = -18$
$(6.8\ \text{k}\Omega + 3.3\ \text{k}\Omega)I_3 - 6.8\ \text{k}\Omega I_2 = -3$

$\mathbf{I_1 = 1.21\ mA,\ I_2 = -0.48\ mA,\ I_3 = -0.62\ mA}$

36. **a.** Write the nodal equations for the networks in Fig. 8.126.
b. Using determinants, solve for the nodal voltages.
c. Determine the magnitude and polarity of the voltage across each resistor.



(I)



(II)

Solution:

(I): V_1 V_2

$$V_1\left[\frac{1}{3}+\frac{1}{6}+\frac{1}{4}\right]-\frac{1}{4}V_2 = -5-3$$

$$V_2\left[\frac{1}{8}+\frac{1}{4}\right]-\frac{1}{4}V_1 = 3-4$$

$V_1 = \mathbf{-14.86\ V}$, $V_2 = \mathbf{-12.57\ V}$

$V_{R_1} = V_{R_4} = \mathbf{-14.86\ V}$

$V_{R_2} = \mathbf{-12.57\ V}$

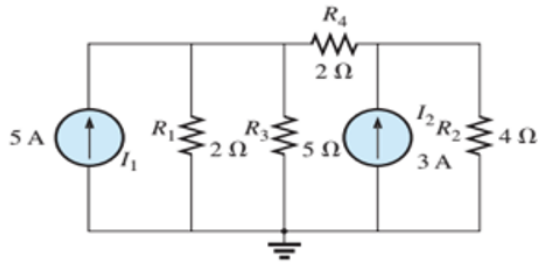
$^{+}V_{R_5}{}^{-} = 12\ \text{V} + 12.57\ \text{V} - 14.86\ \text{V} = \mathbf{9.71\ V}$

(II): I_1, I_3, I_2

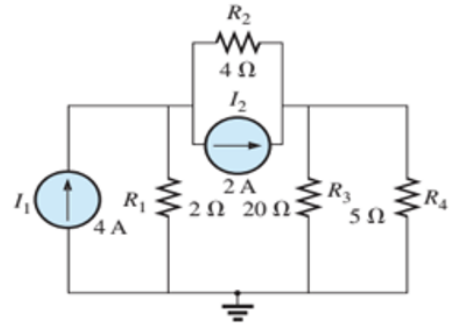
$$(4\,\Omega+4\,\Omega+3\,\Omega)I_1 - 3\,\Omega\, I_2 - 4\,\Omega\, I_3 = 16-12$$
$$(4\,\Omega+3\,\Omega+10\,\Omega)I_2 - 3I_1 - 4\,\Omega\, I_3 = 12-15$$
$$(4\,\Omega+4\,\Omega+7\,\Omega)I_3 - 4I_1 - 4I_2 = -16$$

$I_1 = \mathbf{-0.24\ A}$, $I_2 = \mathbf{-0.52\ A}$, $I_3 = \mathbf{-1.28\ A}$

35. Write the nodal equations for the networks in Fig. 8.125. Using determinants, solve for the nodal voltages. Is symmetry present?



(a)



(b)

a.

$$V_1\left[\frac{1}{2}+\frac{1}{5}+\frac{1}{2}\right]-\frac{1}{2}V_2=5$$

$$V_2\left[\frac{1}{2}+\frac{1}{4}\right]-\frac{1}{2}V_1=3$$

Symmetry is present

$V_1 = \mathbf{8.08\ V}$
$V_2 = \mathbf{9.39\ V}$

b.

$$V_1\left[\frac{1}{2}+\frac{1}{4}\right]-\frac{1}{4}V_2=4-2$$

$$V_2\left[\frac{1}{4}+\frac{1}{20}+\frac{1}{5}\right]-\frac{1}{4}V_1=2$$

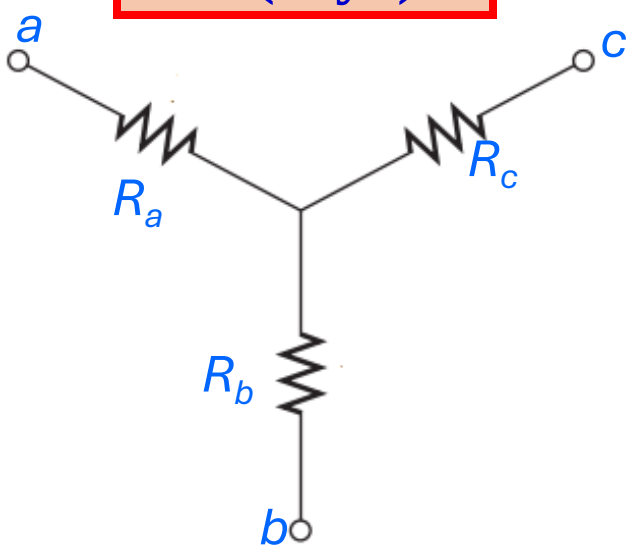
Symmetry is present

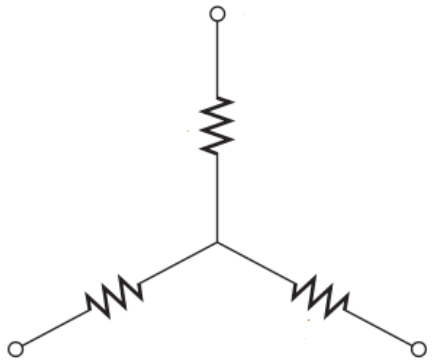
$V_1 = \mathbf{4.80\ V}$
$V_2 = \mathbf{6.40\ V}$

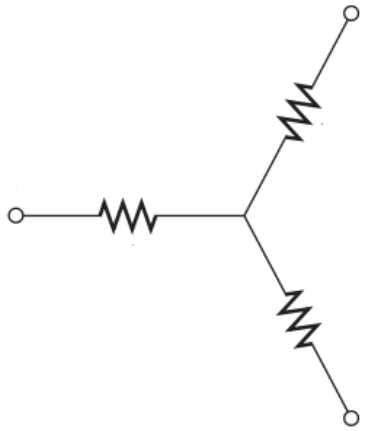
8.12 Y–Δ (T–Π) and Δ–Y (Π–T) CONVERSIONS

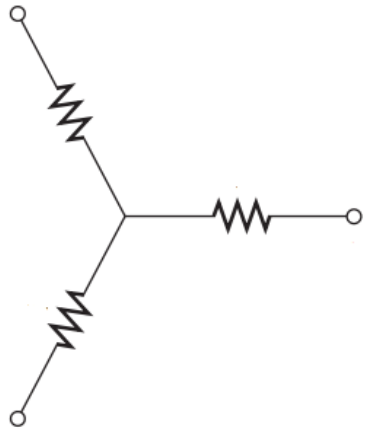
Circuit configurations are often encountered in which the resistors do **not appear to be in series or parallel**. Under these conditions, it may be necessary to convert the circuit from one form to another [*i.e.* convert from **Y/T to Δ/Π or Δ/Π to Y/T**] to solve for any unknown quantities

Y (Wye)

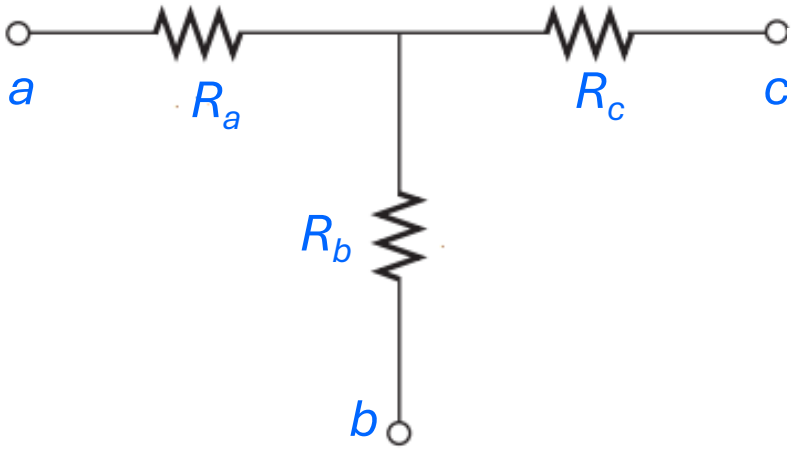


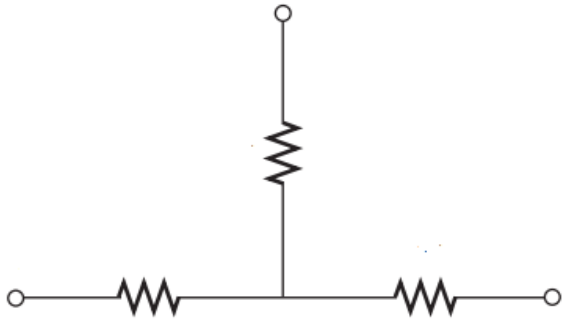


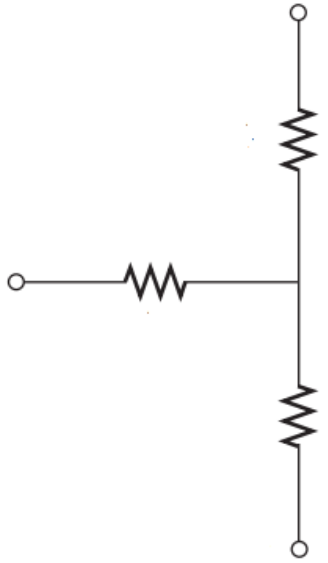


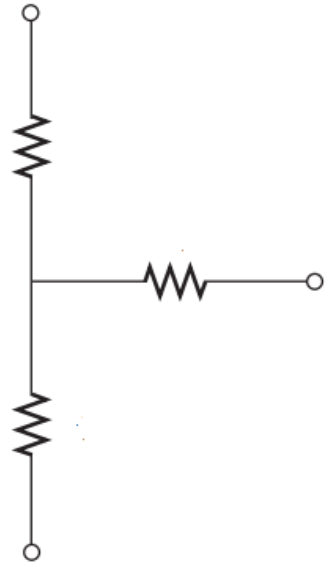


T (Tee)

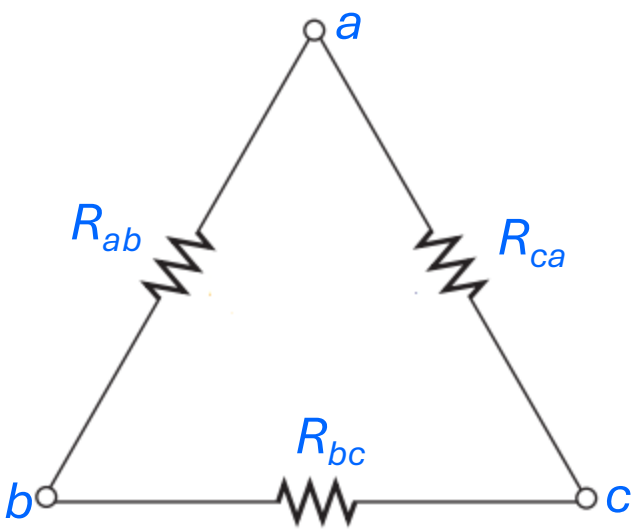


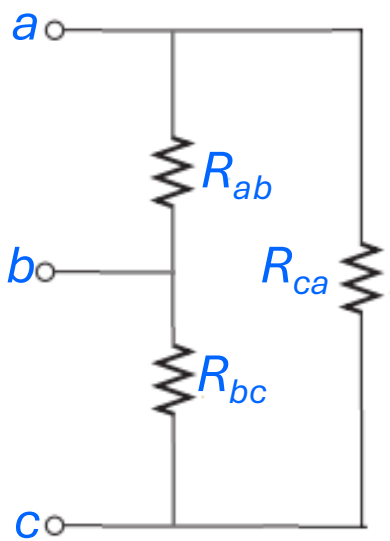


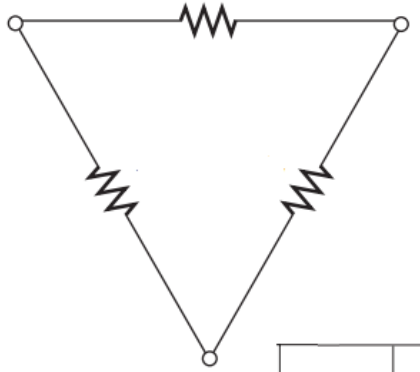


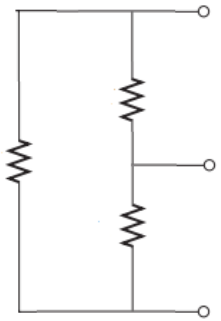


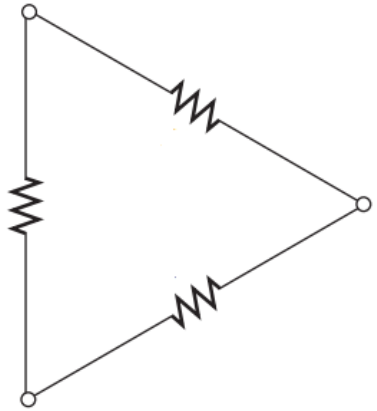
Δ (Delta)

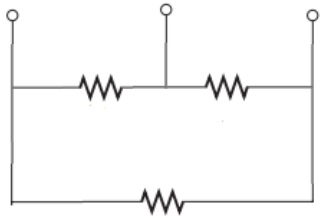


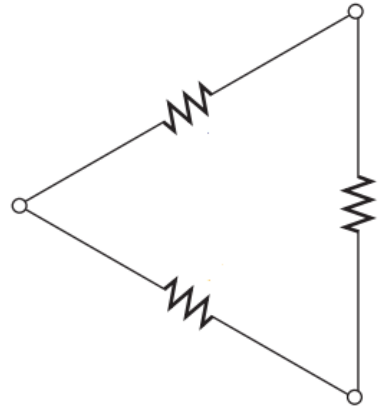


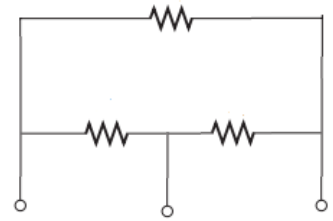




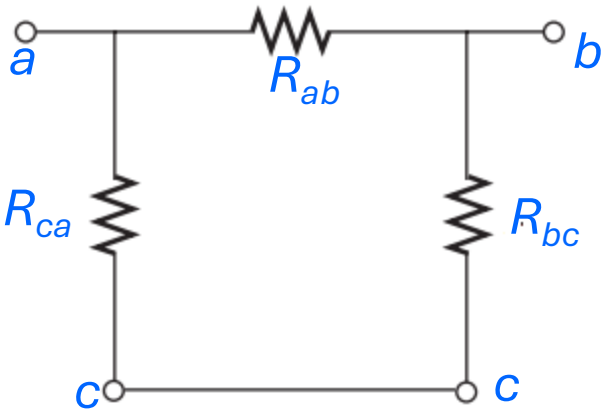


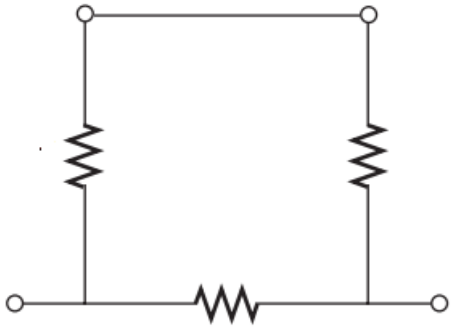


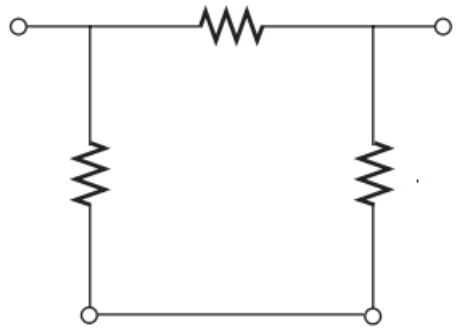


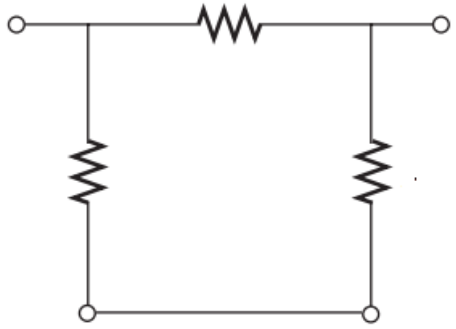


Π (Pai)

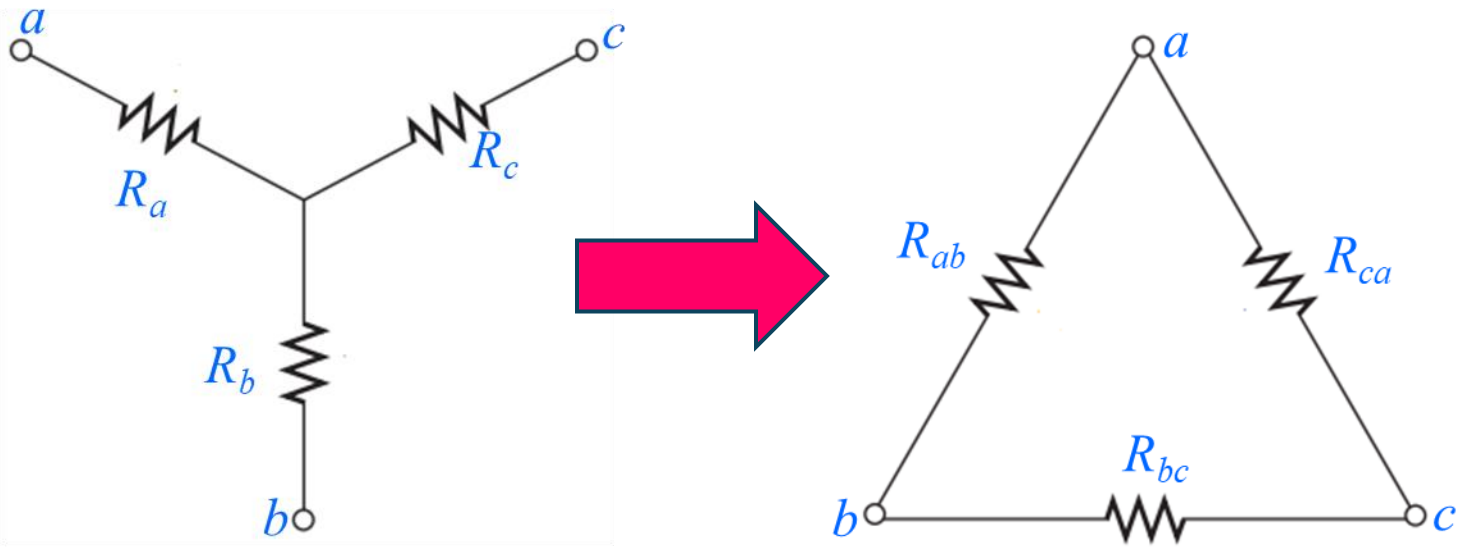








Conversion from Y (T) to Δ (Π)



Here, R_a, R_b and R_c are known.

Calculate R_{ab}, R_{bc} and R_{ca}

$$R_{abc} = R_aR_b + R_bR_c + R_cR_a$$

$$R_{ab} = \frac{R_{abc}}{R_c} = \frac{R_aR_b + R_bR_c + R_cR_a}{R_c}$$

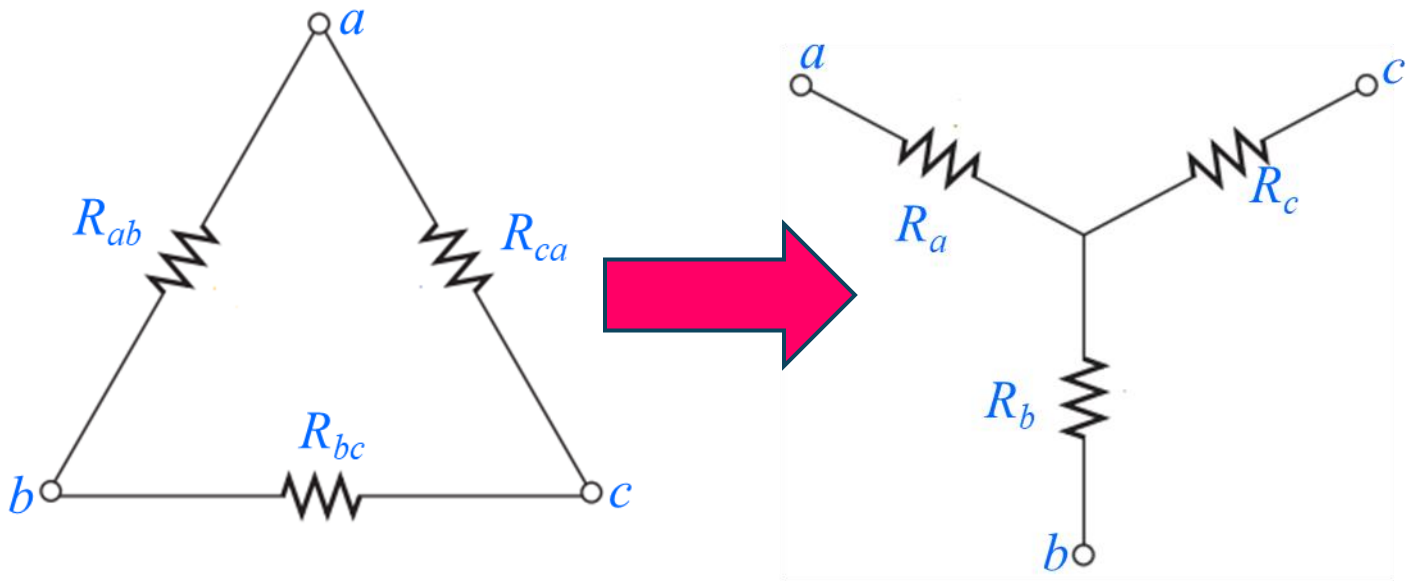
$$R_{bc} = \frac{R_{abc}}{R_a} = \frac{R_aR_b + R_bR_c + R_cR_a}{R_a}$$

$$R_{ca} = \frac{R_{abc}}{R_b} = \frac{R_aR_b + R_bR_c + R_cR_a}{R_b}$$

If $R_a = R_b = R_c = R_Y$ then $R_{ab} = R_{bc} = R_{ca} = R_\Delta = 3R_Y$

For derivation of these equations go through Eq. (8.3a) to (8.4 c)

Conversion from Δ (Π) to Y (T)



Here, R_{ab}, R_{bc} and R_{ca} are known.

Calculate R_a, R_b and R_c

$$R_{abc} = R_{ab} + R_{bc} + R_{ca}$$

$$R_a = \frac{R_{ab}R_{ca}}{R_{abc}} = \frac{R_{ab}R_{ca}}{R_{ab} + R_{bc} + R_{ca}}$$

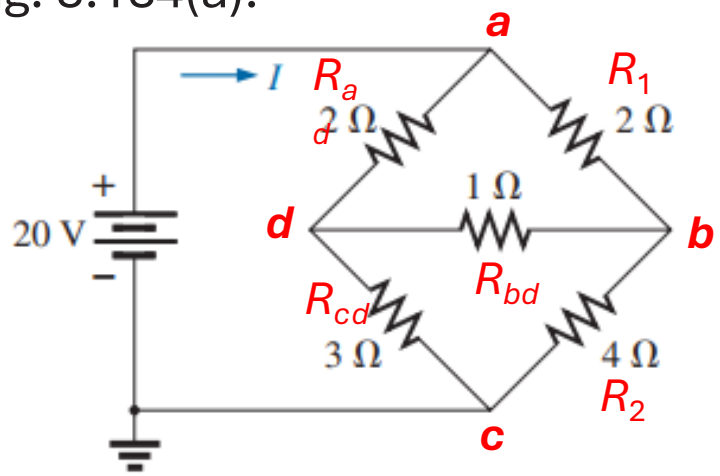
$$R_b = \frac{R_{bc}R_{ab}}{R_{abc}} = \frac{R_{bc}R_{ab}}{R_{ab} + R_{bc} + R_{ca}}$$

$$R_c = \frac{R_{ca}R_{bc}}{R_{abc}} = \frac{R_{ca}R_{bc}}{R_{ab} + R_{bc} + R_{ca}}$$

If $R_{ab} = R_{bc} = R_{ca} = R_\Delta$ then $R_{ab} = R_{bc} = R_{ca} = R_Y = \frac{R_\Delta}{3}$

For derivation of these equations go through Eq. (8.5a) to (8.5c)

Problem 51(a) [P. 342]. Using a Δ-Y or Y-Δ conversion, find the current *I* in each of the networks in Fig. 8.134(a).



Solution: First, marked four nodes (*a*, *b*, *c*, and *d*) in the circuit.

Let, $R_{ad} = 2\ \Omega$, $R_{bd} = 1\ \Omega$, $R_{cd} = 3\ \Omega$, $R_1 = 2\ \Omega$, and $R_2 = 4\ \Omega$

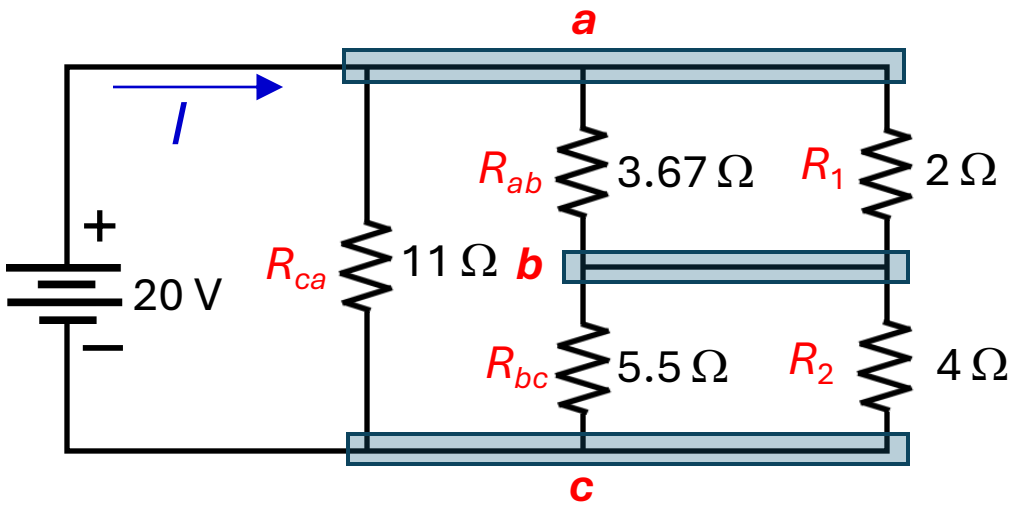
Here, R_{ad}, R_{bd}, and R_{cd} made Y connection where *d* is common point. So, this Y connection is going to convert Δ connection.

$$R_{abc} = R_{ad}R_{bd} + R_{bd}R_{cd} + R_{cd}R_{ad}$$
$$= (2\Omega)(1\Omega) + (1\Omega)(3\Omega) + (3\Omega)(2\Omega)$$
$$= 2\Omega + 3\Omega + 6\Omega = 11\Omega$$

$$R_{ab} = \frac{R_{abc}}{R_{cd}} = \frac{11\Omega}{3\Omega} = 3.67\Omega$$

$$R_{bc} = \frac{R_{abc}}{R_{ad}} = \frac{11\Omega}{2\Omega} = 5.5\Omega$$

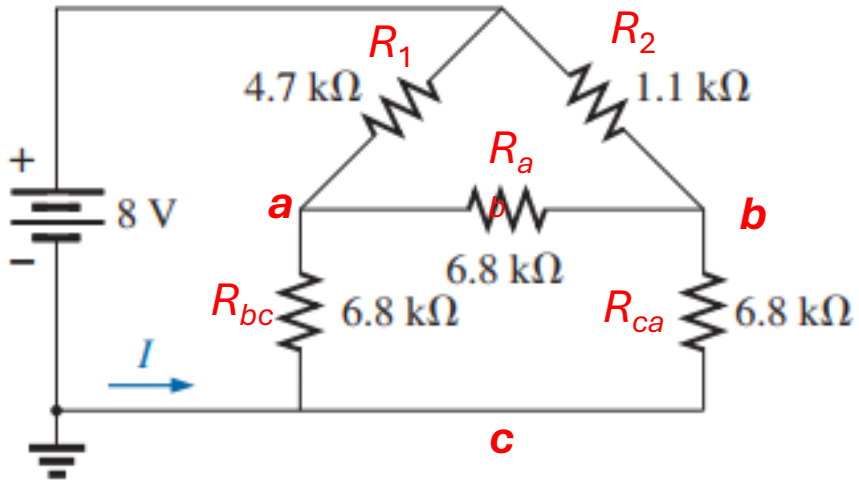
$$R_{ca} = \frac{R_{abc}}{R_{bd}} = \frac{11\Omega}{1\Omega} = 11\Omega$$



$$R_T = (R_{ca}) // \{(R_{ab} // R_1) + (R_{bc} // R_2)\}$$
$$= (11\Omega) // \{(3.67\Omega // 2\Omega) + (5.5\Omega // 4\Omega)\}$$
$$= (11\Omega) // \{1.29\Omega + 2.32\Omega\}$$
$$= (11\Omega) // \{3.61\Omega\}$$
$$= 2.72\ \Omega$$

$$I = \frac{E}{R_T} = \frac{20\,\text{V}}{2.72\,\Omega} = \mathbf{7.35\,V}$$

Problem 51(b) [P. 342]. Using a Δ-Y or Y-Δ conversion, find the current *I* in each of the networks in Fig. 8.134(b).



Solution: First, marked four nodes (*a*, *b*, and *c*) in the circuit.

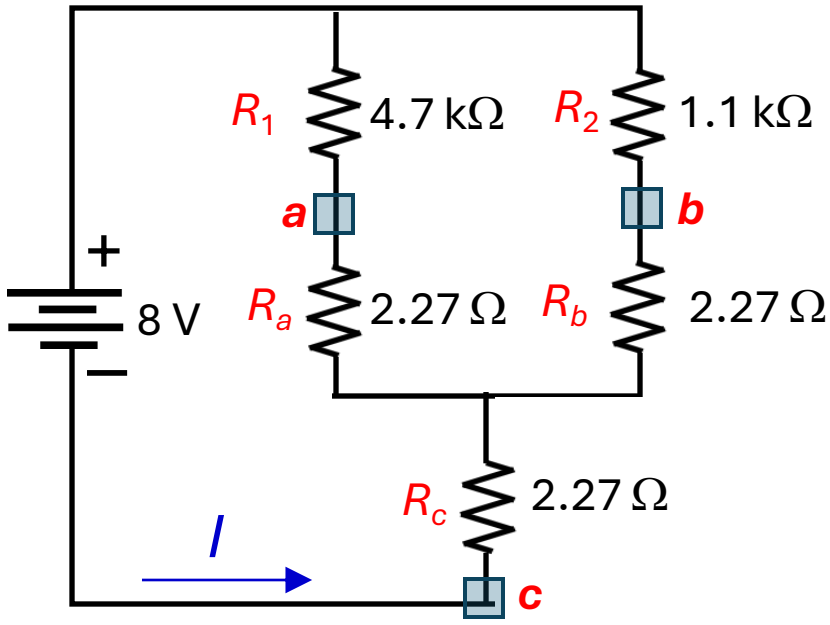
Let, R_1=4.7 kΩ, R_2=1.1 kΩ, R_{ab}=R_{bc}=R_{ca}= 6.8 kΩ

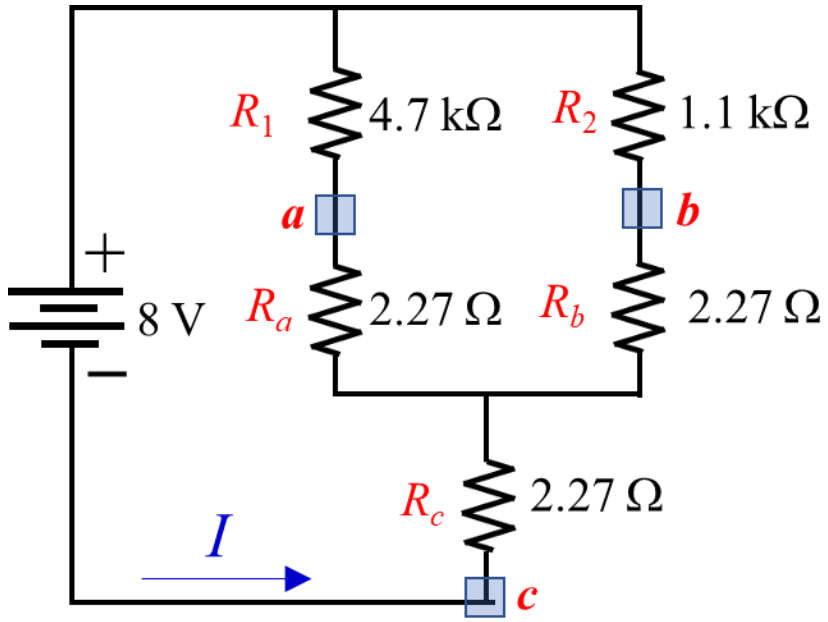
Here, R_{ab}, R_{bc}, and R_{ca} made Δ connection. So, this Δ connection is going to convert Y connection.

Here, $R_{ab} = R_{bc} = R_{ca} = R_\Delta = 6.8\ \text{k}\Omega$

so, $R_a = R_b = R_c = R_Y = \dfrac{R_\Delta}{3}$

$$= \frac{6.8\ \text{k}\Omega}{3} = 2.27\ \text{k}\Omega$$





$$R_T = [(R_1 + R_a)//(R_2 + R_b)] + R_c$$
$$= [(4.7\text{k}\Omega + 2.27\text{k}\Omega)//(1.1\text{k}\Omega + 2.27\text{k}\Omega)] + 2.27\text{k}\Omega$$
$$= [(6.97\text{k}\Omega)//(3.37\text{k}\Omega)] + 2.27\text{k}\Omega$$
$$= 2.27\text{k}\Omega + 2.27\text{k}\Omega$$
$$= 4.54\text{k}\Omega$$

$$I = \frac{E}{R_T} = \frac{8\,\text{V}}{4.54\ \text{k}\Omega} = \mathbf{1.76\ mA}$$

Practice Book [Ch 8] Problem: 51 ~ 56

8.11 BRIDGE NETWORKS

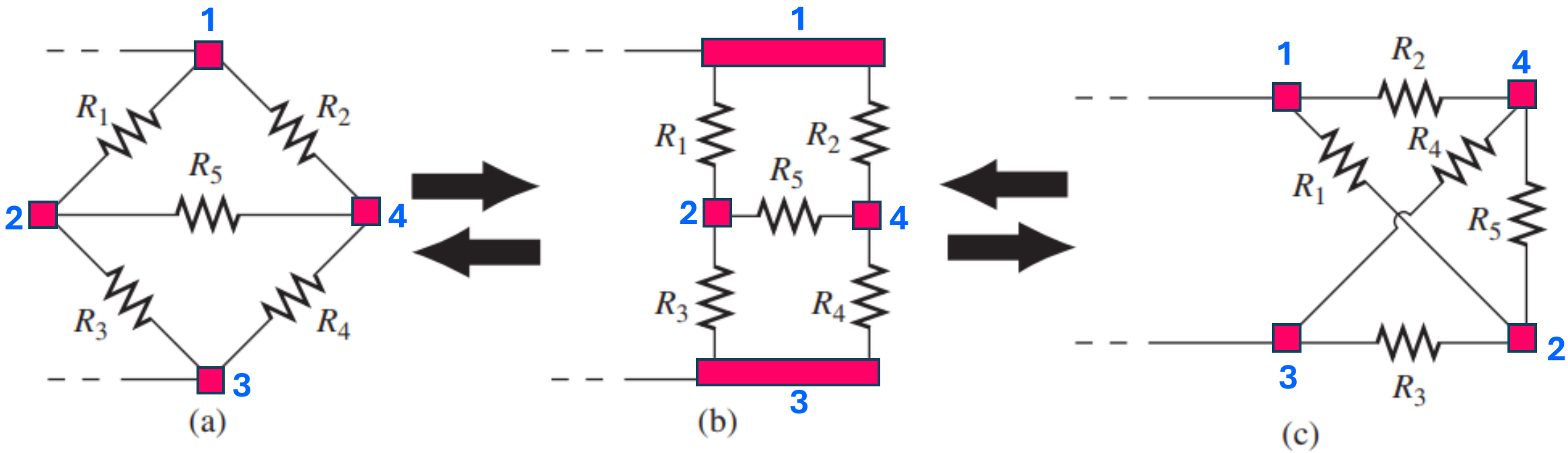
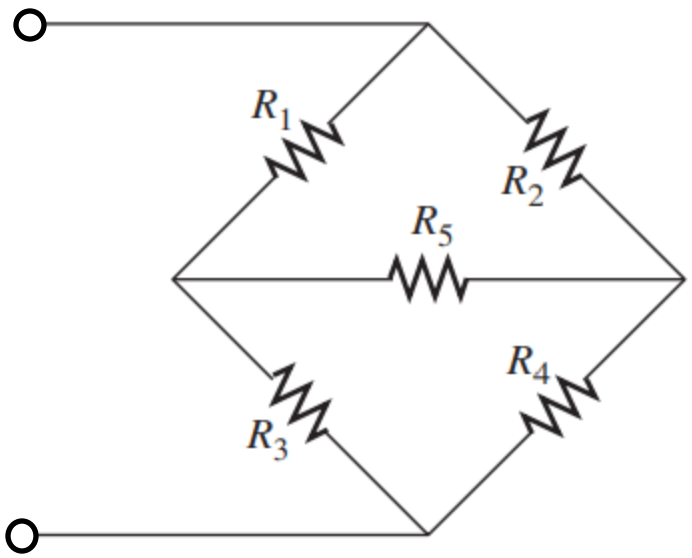


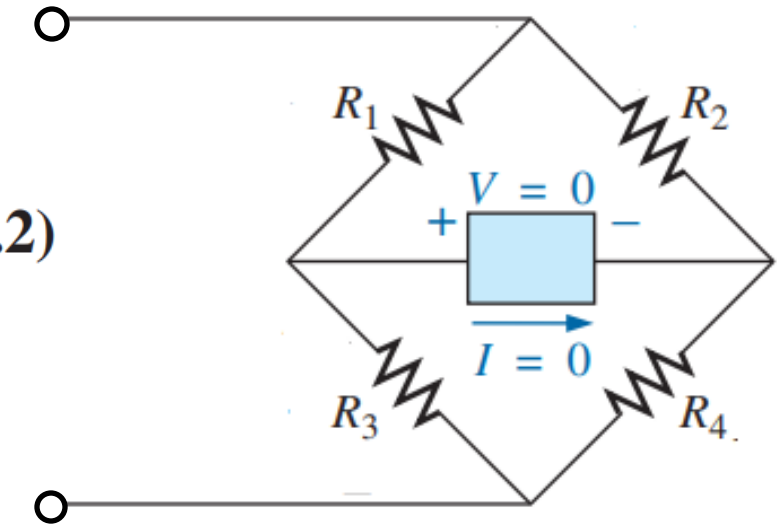
FIG. 8.69 *Various formats for a bridge network.*

Balanced Bridge Network



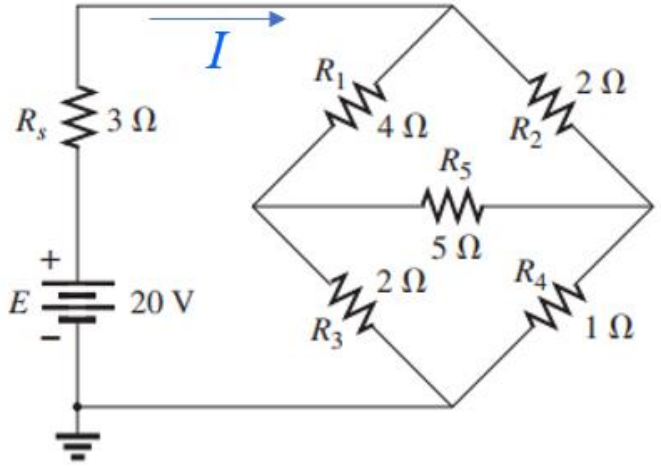
If the ratio of R_1 to R_3 is equal to that of R_2 to R_4, the bridge is balanced, and $I = 0$ A or $V = 0$ V.

$$\frac{R_1}{R_3} = \frac{R_2}{R_4} \qquad (8.2)$$

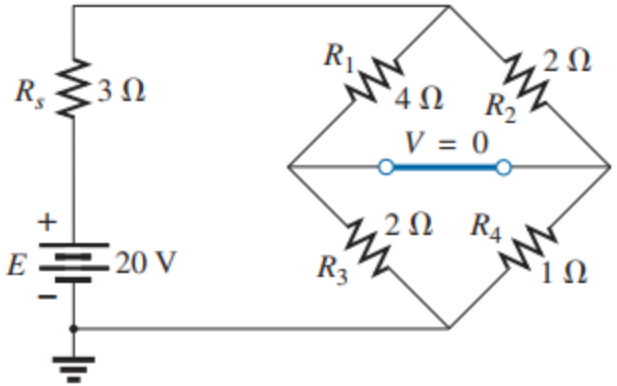


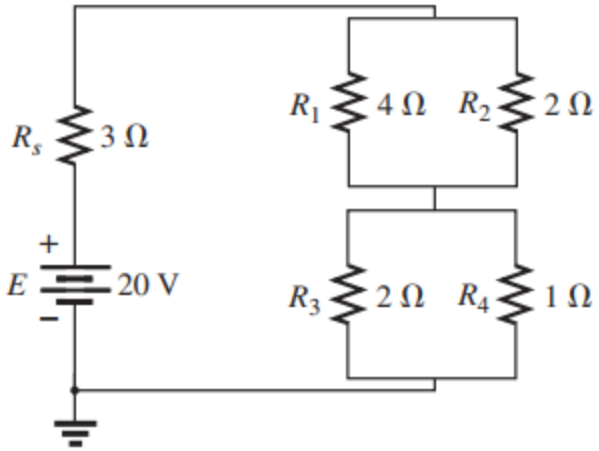
If this condition is not satisfied, to solve the bridge circuit use the other techniques.

Example 8.11.1: Find the current *I* for the following network.



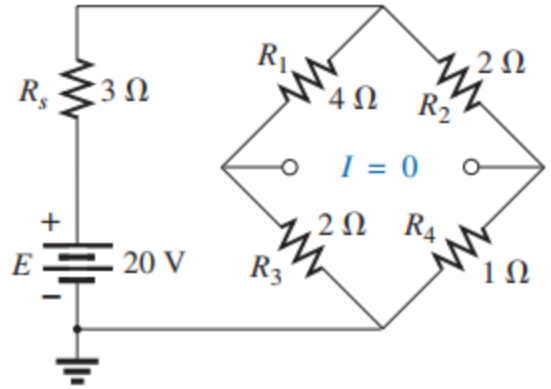
Solution: Here: $\frac{R_1}{R_3} = \frac{4\Omega}{2\Omega} = 2$ and $\frac{R_2}{R_4} = \frac{2\Omega}{1\Omega} = 2$ so $\frac{R_1}{R_3} = \frac{R_2}{R_4}$

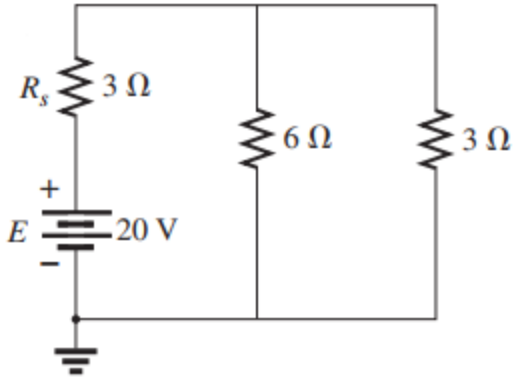




$$R_T = 3\Omega + (4\Omega // 2\Omega) + (2\Omega // 1\Omega) = 5\Omega$$

$$I = \frac{E}{R_T} = \frac{20\,\text{V}}{5\Omega} = \mathbf{4\,A}$$





$$R_T = 3\Omega + (6\Omega // 3\Omega) = 5\Omega$$

$$I = \frac{E}{R_T} = \frac{20\,\text{V}}{5\Omega} = \mathbf{4\,A}$$

Chapter 9

SUPERPOSITION THEOREM

Find the solution to networks with two or more sources that are not in series or parallel.

Does not require the use of a mathematical technique such as determinants to find the required voltages or currents.

The superposition theorem states the following:

The current through, or voltage across, an element in a linear bilateral network is equal to the algebraic sum of the currents or voltages produced independently by each source.

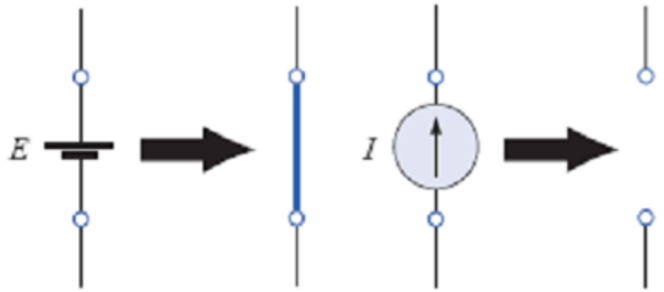
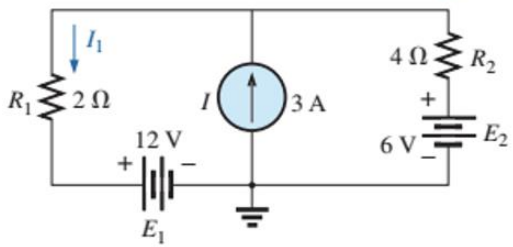


FIG. 9.1 *Removing the effects of ideal sources.*

9.2 SUPERPOSITION THEOREM

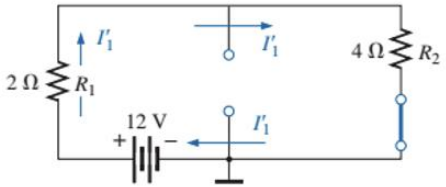
EXAMPLE 9.5 Find the current through the 2 Ω resistor of the network in Fig. 9.18. The presence of three sources results in three different networks to be analyzed.



Solution: **Step 1:**

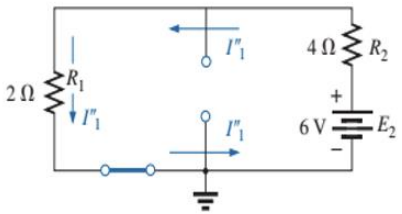
Considering the effect of the 12 V source (Fig. 9.19):

$$I'_1 = \frac{E_1}{R_1 + R_2} = \frac{12\text{ V}}{2\,\Omega + 4\,\Omega} = \frac{12\text{ V}}{6\,\Omega} = 2\text{ A}$$



Step 2:

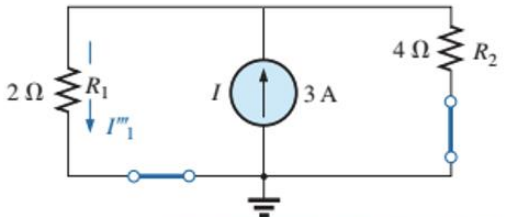
Considering the effect of the 6 V source (Fig. 9.20):



$$= \frac{E_2}{R_1 + R_2} = \frac{6\text{ V}}{2\,\Omega + 4\,\Omega} = \frac{6\text{ V}}{6\,\Omega} = 1\text{ A}$$

Step 3:

$$I'''_1 = \frac{R_2 I}{R_1 + R_2} = \frac{(4\,\Omega)(3\text{ A})}{2\,\Omega + 4\,\Omega} = \frac{12\text{ A}}{6} = 2\text{ A}$$

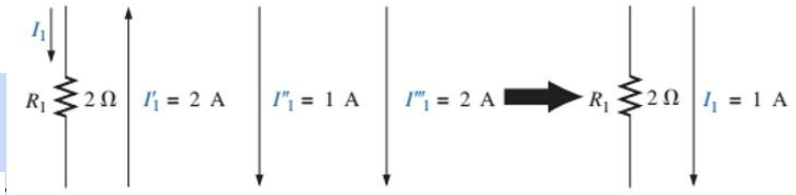


Step 4:

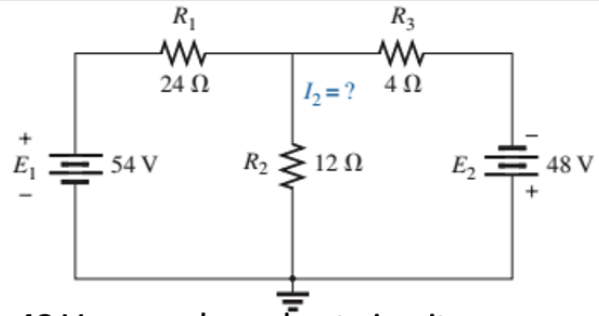
The total current through the 2 Ω resistor appears in Fig. 9.22 and

Same direction as I_1 in Fig. 9.18 — Opposite direction to I_1 in Fig. 9.18

$$I_1 = \overbrace{I''_1 + I'''_1}^{\text{Same direction as } I_1 \text{ in Fig. 9.18}} \overbrace{-\,I'_1}^{\text{Opposite direction to } I_1 \text{ in Fig. 9.18}}$$
$$= 1\text{ A} + 2\text{ A} - 2\text{ A} = 1\text{ A}$$



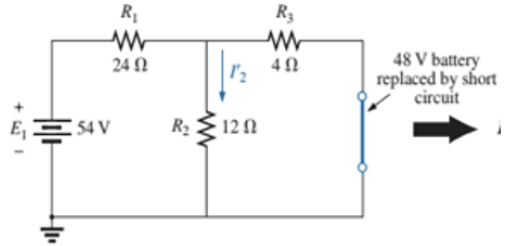
EXAMPLE 9.2 Using the superposition theorem, determine the current through the 12 Ω resistor in Fig. 9.5. Note that this is a two-source network of the type examined in the previous chapter when we applied branch-current analysis and mesh analysis.



Solution:

Step 1:

Considering the effects of the 54 V source requires replacing the 48 V source by a short-circuit equivalent.

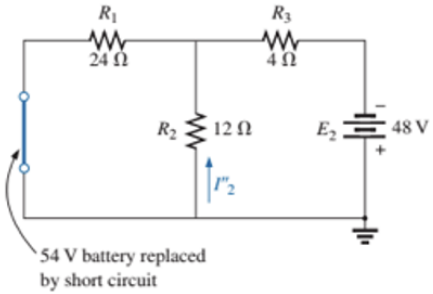


$$R_T = R_1 + R_2 \| R_3 = 24\ \Omega + 12\ \Omega \| 4\ \Omega = 24\ \Omega + 3\ \Omega = 27\ \Omega$$

$$I_s = \frac{E_1}{R_T} = \frac{54\ \text{V}}{27\ \Omega} = 2\ \text{A} \qquad I'_2 = \frac{R_3 I_s}{R_3 + R_2} = \frac{(4\ \Omega)(2\ \text{A})}{4\ \Omega + 12\ \Omega} = 0.5\ \text{A}$$

Step 2:

If we now replace the 54 V source by a short-circuit equivalent, the network in Fig. 9.7 results.

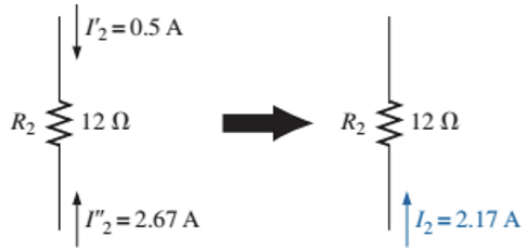


$$R_T = R_3 + R_2 \| R_1 = 4\ \Omega + 12\ \Omega \| 24\ \Omega = 4\ \Omega + 8\ \Omega = 12\ \Omega$$

$$I_s = \frac{E_2}{R_T} = \frac{48\ \text{V}}{12\ \Omega} = 4\ \text{A} \qquad I''_2 = \frac{R_1(I_s)}{R_1 + R_2} = \frac{(24\ \Omega)(4\ \text{A})}{24\ \Omega + 12\ \Omega} = 2.67\ \text{A}$$

Step 3:

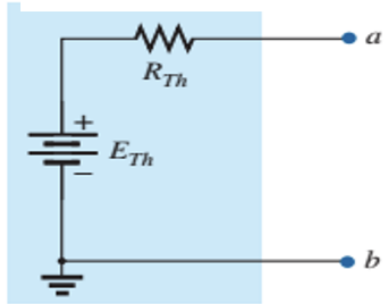
$$I_2 = I''_2 - I'_2 = 2.67\ \text{A} - 0.5\ \text{A} = \mathbf{2.17\ A}$$



9.3 THÉVENIN'S THEOREM

Any two-terminal dc network can be replaced by an equivalent circuit consisting solely of a voltage source and a series resistor as shown in the figure. This is called Thevenin Equivalent Circuit.

Thévenin equivalent circuit.



Steps:

1. Remove that portion of the network where the Thévenin equivalent circuit is found. In this circuit the load resistor R_L be temporarily removed from the network. Marked the terminals (a and b) across which thevenin is going to be applied.

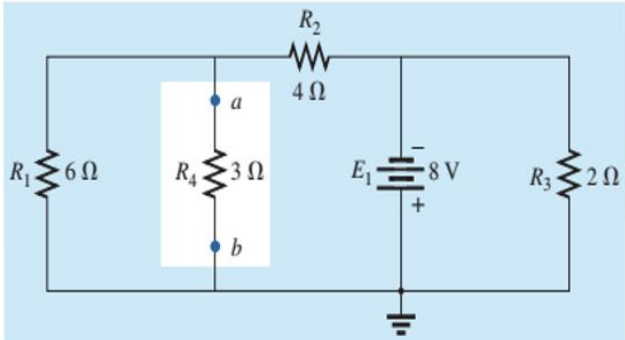
2. Calculate R_{Th} by first setting all sources to zero (voltage sources are replaced by short circuits, and current sources by open circuits).

3. Consider an imaginary current is entering through the marked terminal 'a'. Calculate R_{Th}.

4. Calculate E_{Th} by first returning all sources to their original position and finding the open-circuit voltage (V_{ab}) between the marked terminals a and b.

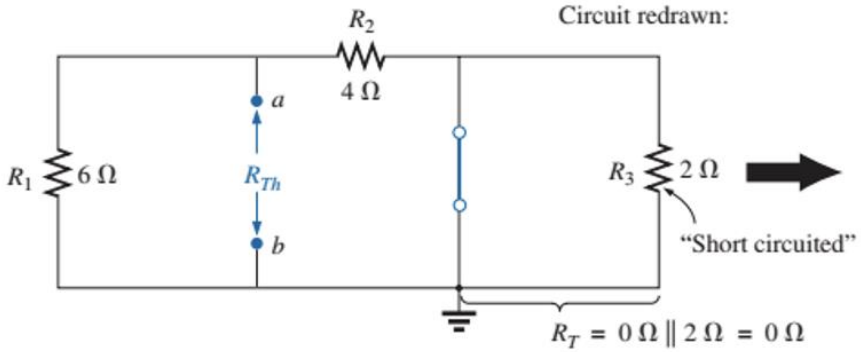
5. Draw the Thévenin equivalent circuit.

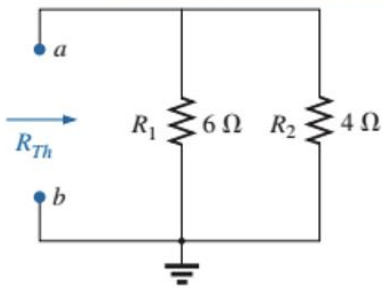
EXAMPLE 9.8 Find the Thévenin equivalent circuit for the network in the shaded area of the network in Fig. 9.37. Note in this example that there is no need for the section of the network to be preserved to be at the "end" of the configuration.



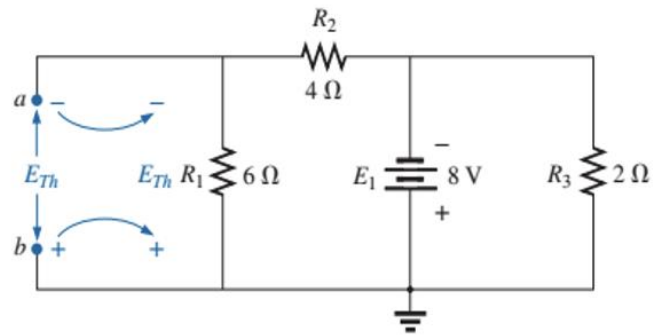
Solution:

Circuit redrawn:



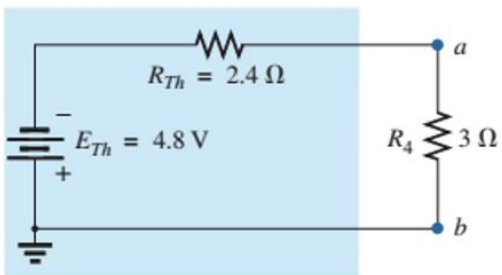


$$R_{Th} = R_1 \| R_2 = \frac{(6\,\Omega)(4\,\Omega)}{6\,\Omega + 4\,\Omega} = \frac{24\,\Omega}{10} = \mathbf{2.4\,\Omega}$$

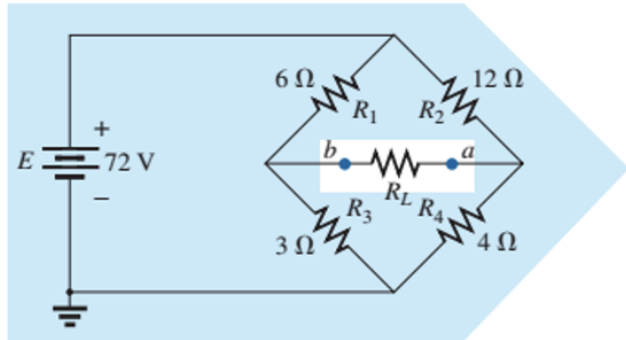


$$E_{Th} = \frac{R_1 E_1}{R_1 + R_2} = \frac{(6\,\Omega)(8\,\text{V})}{6\,\Omega + 4\,\Omega} = \frac{48\,\text{V}}{10} = \mathbf{4.8\,V}$$

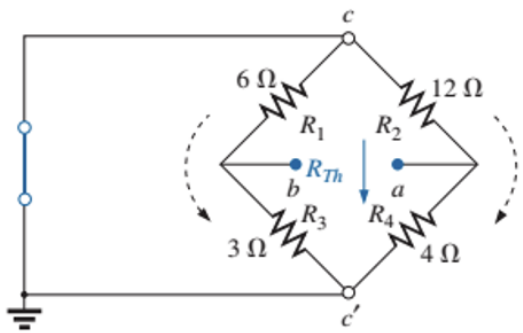
Resultant Thevenin Equivalent Circuit:

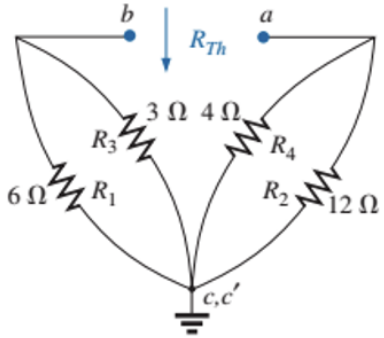


EXAMPLE 9.9 Find the Thévenin equivalent circuit for the network in the shaded area of the bridge network in Fig. 9.43.



Solution:

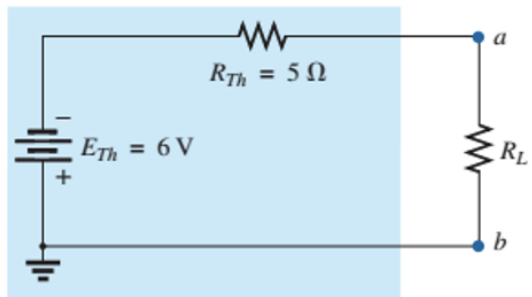


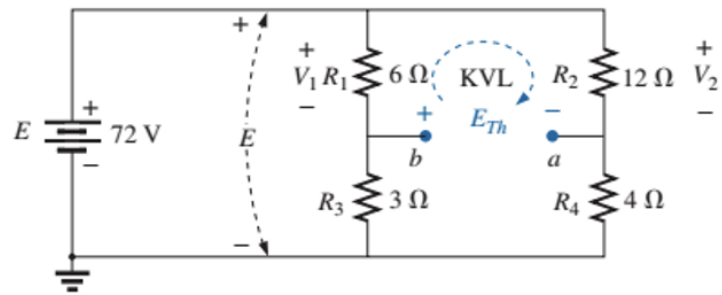


$$R_{Th} = R_{a-b} = R_1 \parallel R_3 + R_2 \parallel R_4$$
$$= 6\,\Omega \parallel 3\,\Omega + 4\,\Omega \parallel 12\,\Omega$$
$$= 2\,\Omega + 3\,\Omega = \mathbf{5\,\Omega}$$

$$V_1 = \frac{R_1 E}{R_1 + R_3} = \frac{(6\,\Omega)(72\text{ V})}{6\,\Omega + 3\,\Omega} = \frac{432\text{ V}}{9} = 48\text{ V}$$

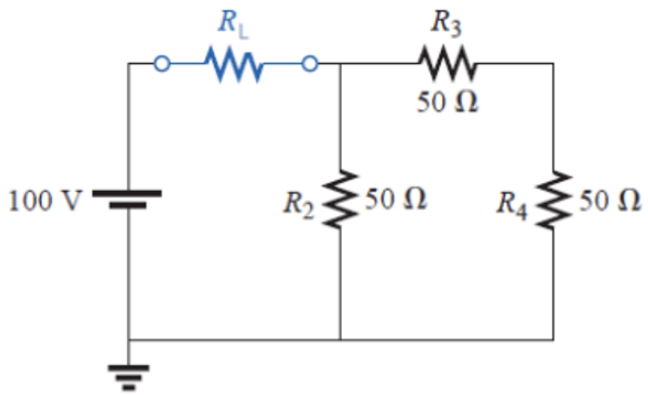
$$V_2 = \frac{R_2 E}{R_2 + R_4} = \frac{(12\,\Omega)(72\text{ V})}{12\,\Omega + 4\,\Omega} = \frac{864\text{ V}}{16} = 54\text{ V}$$

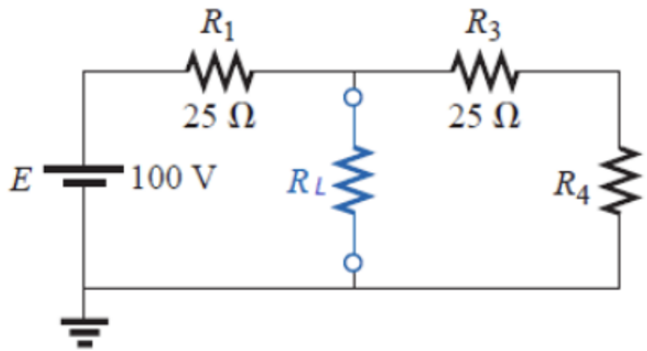




$$\Sigma_C V = +E_{Th} + V_1 - V_2 = 0$$
$$E_{Th} = V_2 - V_1 = 54\text{ V} - 48\text{ V} = \mathbf{6\text{ V}}$$

Class Practice





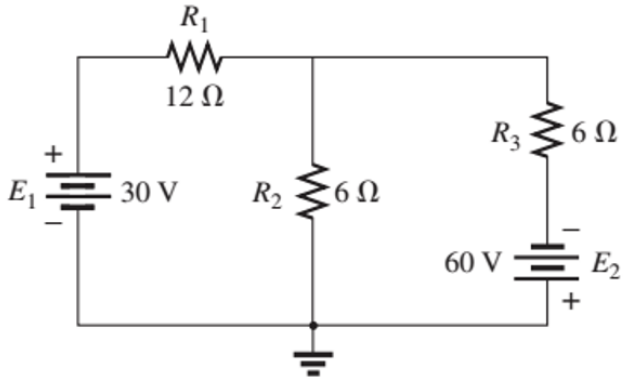
Exercise Problems

1. a. Using superposition, find the current through each resistor of the network in Fig. 9.119.

b. Find the power delivered to R_1 for each source.

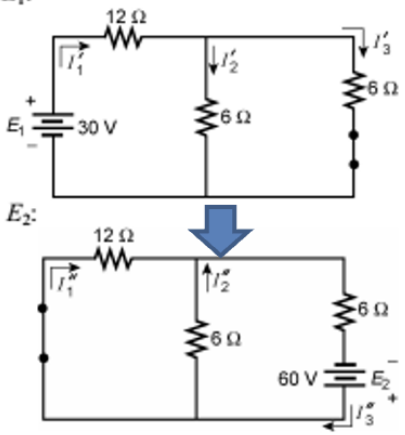
c. Find the power delivered to R_1 using the total current through R_1.

d. Does superposition apply to power effects? Explain.



Solution:

a. E_1:



$$I'_1 = \frac{30\text{ V}}{12\,\Omega + 6\,\Omega \| 6\,\Omega} = \frac{30\text{ V}}{12\,\Omega + 3\,\Omega} = 2\text{ A}$$

$$I'_2 = I'_3 = \frac{I'_1}{2} = 1\text{ A}$$

$$I''_3 = \frac{60\text{ V}}{6\,\Omega + 6\,\Omega \| 12\,\Omega} = \frac{60\text{ V}}{6\,\Omega + 4\,\Omega} = 6\text{ A}$$

$$I''_1 = \frac{6\,\Omega(I''_3)}{6\,\Omega + 12\,\Omega} = 2\text{ A}$$

$$I''_2 = \frac{12\,\Omega(I''_3)}{12\,\Omega + 6\,\Omega} = 4\text{ A}$$

$I_1 = I'_1 + I''_1 = 2\text{ A} + 2\text{ A} = \mathbf{4\text{ A}}$ (dir. of I'_1)

$I_2 = I''_2 - I'_2 = 4\text{ A} - 1\text{ A} = \mathbf{3\text{ A}}$ (dir. of I''_2)

$I_3 = I'_3 + I''_3 = 1\text{ A} + 6\text{ A} = \mathbf{7\text{ A}}$ (dir. of I'_3)

b. E_1: $P'_1 = I'^2_1 R_1 = (2\text{ A})^2\,12\,\Omega = \mathbf{48\text{ W}}$

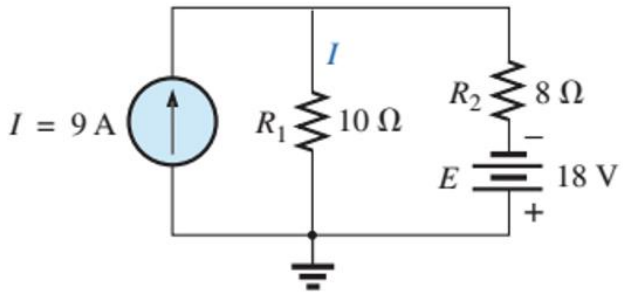
E_2: $P''_1 = I''^2_1 R_1 \text{ I''} R_1 = (2\text{ A})^2\,12\,\Omega = \mathbf{48\text{ W}}$

c. $P_1 = I_1^2 R_1 = (4\text{ A})^2\,12\,\Omega = \mathbf{192\text{ W}}$

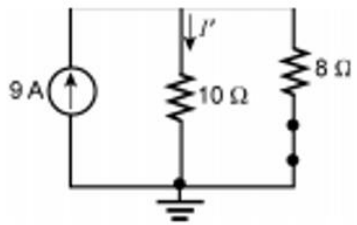
d. $P'_1 + P''_1 = 48\text{ W} + 48\text{ W} = 96\text{ W} \neq 192\text{ W} = P_1$

2. Using superposition, find the current I through the 10 Ω resistor for the network in Fig. 9.120.

Solution:

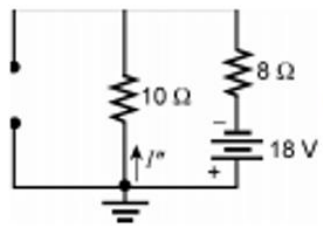


I:



$$I' = \frac{8\,\Omega(9\text{ A})}{8\,\Omega + 10\,\Omega} = 4\text{ A}$$

E:

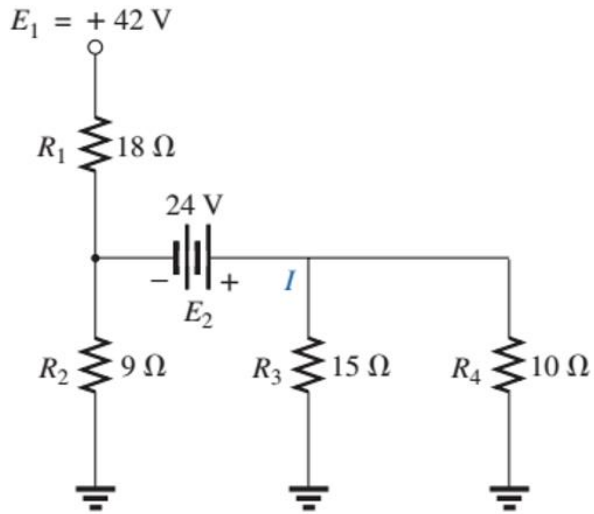


$$I'' = \frac{18\text{ V}}{10\,\Omega + 8\,\Omega} = 1\text{ A}$$



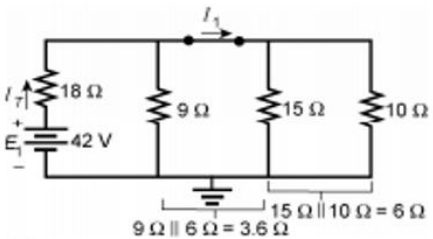
$$I = I' - I'' = 4\text{ A} - 1\text{ A} = \mathbf{3\text{ A (dir of } I')}$$

3. Using superposition, find the current I through the 24 V source in Fig. 9.121.



Solution:

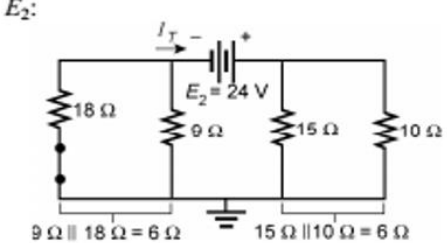
E_1:



$$I_T = \frac{42\text{ V}}{18\Omega + 3.6\Omega} = 1.944\text{ A}$$

$$I_1 = \frac{9\Omega(I_T)}{9\Omega + 6\Omega} = \frac{9\Omega(1.944\text{ A})}{15\Omega} = 1.17\text{ A}$$

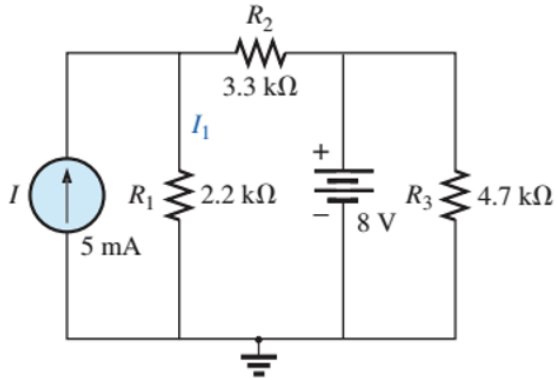
E_2:



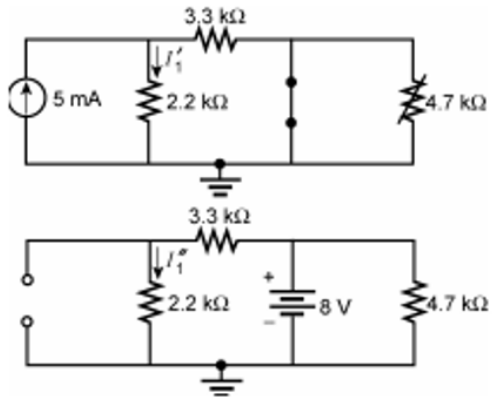
$$I_T = \frac{E_2}{R_T} = \frac{24\text{ V}}{12\Omega} = 2\text{ A}$$

$I_{24V} = I_T + I_1 = 2\text{ A} + 1.17\text{ A} = \mathbf{3.17\text{ A (dir. of } I_1)}$

*4. Using superposition, find the current through R_1 for the network in Fig. 9.122.



Solution:



$$I_1' = \frac{3.3\,\text{k}\Omega(5\,\text{mA})}{2.2\,\text{k}\Omega + 3.3\,\text{k}\Omega} = \frac{16.5\,\text{mA}}{5.5}$$

$$= 3\,\text{mA}$$

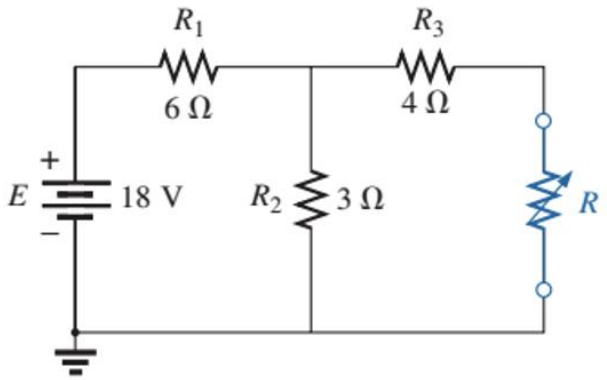
$$I_1'' = \frac{8\,\text{V}}{3.3\,\text{k}\Omega + 2.2\,\text{k}\Omega} = \frac{8\,\text{V}}{5.5\,\text{k}\Omega}$$

$$= 1.45\,\text{mA}$$

$$I_1 = I_1' + I_1'' = 3\,\text{mA} + 1.45\,\text{mA} = \mathbf{4.45\,mA}$$

7. **a.** Find the Thévenin equivalent circuit for the network external to the resistor R in Fig. 9.125.

b. Find the current through R when R is 2 Ω, 30 Ω, and 100 Ω.



Solution:

a. $R_{Th} = R_3 + R_1 \| R_2 = 4\,\Omega + 6\,\Omega \| 3\,\Omega = 4\,\Omega + 2\,\Omega = \mathbf{6\,\Omega}$

$$E_{Th} = \frac{R_2 E}{R_2 + R_1} = \frac{3\,\Omega(18\text{ V})}{3\,\Omega + 6\,\Omega} = 6\text{ V}$$

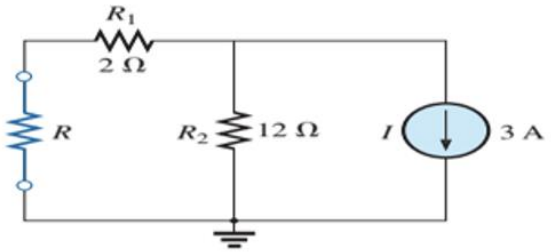
b. $$I_1 = \frac{E_{Th}}{R_{Th} + R} = \frac{6\text{ V}}{6\,\Omega + 2\,\Omega} = \mathbf{0.75\text{ A}}$$

$$I_2 = \frac{6\text{ V}}{6\,\Omega + 30\,\Omega} = \mathbf{166.67\text{ mA}}$$

$$I_3 = \frac{6\text{ V}}{6\,\Omega + 100\,\Omega} = \mathbf{56.60\text{ mA}}$$

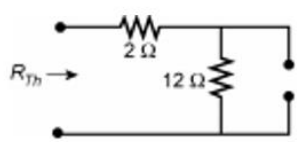
8. **a.** Find the Thévenin equivalent circuit for the network external to the resistor R for the network in Fig. 9.126.

b. Find the power delivered to R when R is 2 Ω and 100 Ω.

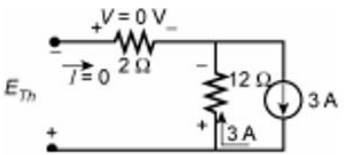


Solution:

a.



$R_{Th} = 2\,\Omega + 12\,\Omega = \mathbf{14\,\Omega}$



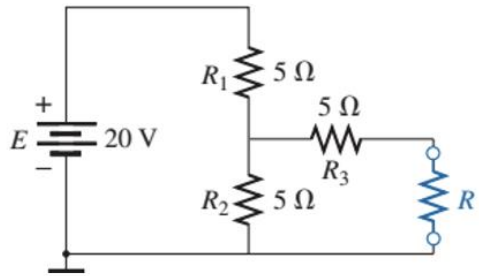
$E_{Th} = IR = (3\text{ A})(12\,\Omega) = \mathbf{36\text{ V}}$

b. $$R = 2\,\Omega: P = \left(\frac{E_{Th}}{R_{Th} + R}\right)^2 R = \left(\frac{36\text{ V}}{14\,\Omega + 2\,\Omega}\right)^2 2\,\Omega = \mathbf{10.13\text{ W}}$$

$$R = 100\,\Omega: P = \left(\frac{36\text{ V}}{14\,\Omega + 100\,\Omega}\right)^2 100\,\Omega = \mathbf{9.97\text{ W}}$$

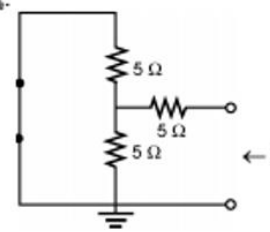
9. **a.** Find the Thévenin equivalent circuit for the network external to the resistor R for the network in Fig. 9.127.

b. Find the power delivered to R when R is 2 Ω and 100 Ω.



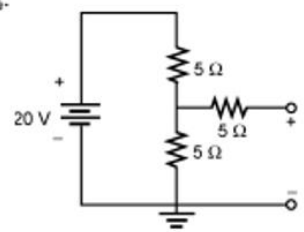
Solution:

a. R_{Th}:



$\leftarrow R_{Th} = 5\,\Omega + 5\,\Omega \parallel 5\,\Omega = \mathbf{7.5\,\Omega}$

E_{Th}:

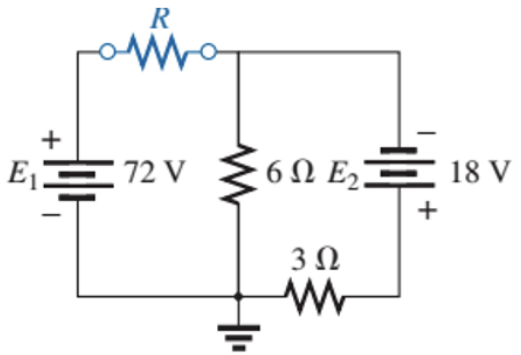


$$E_{Th} = \frac{20\text{ V}}{2} = \mathbf{10\text{ V}}$$

b. $R = 2\,\Omega$: $P = \left(\frac{E_{Th}}{R_{Th}+R}\right)^2 R = \left(\frac{10\text{ V}}{7.5\,\Omega+2\,\Omega}\right)^2 2\,\Omega = \mathbf{2.22\text{ W}}$

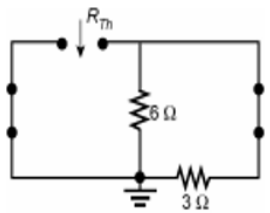
$R = 100\,\Omega$: $P = \left(\frac{10\text{ V}}{7.5\,\Omega+100\,\Omega}\right)^2 100\,\Omega = \mathbf{0.87\text{ W}}$

10. Find the Thévenin equivalent circuit for the network external to the resistor R for the network in Fig. 9.128.



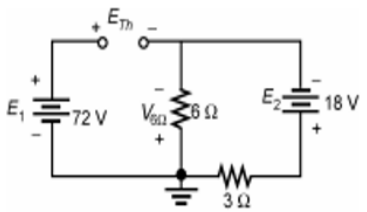
Solution:

R_{Th}:



$$R_{Th} = 6\,\Omega \,\|\, 3\,\Omega = \mathbf{2\,\Omega}$$

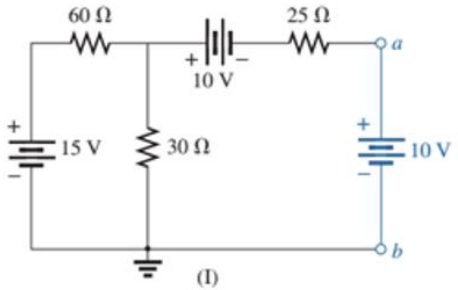
E_{Th}:

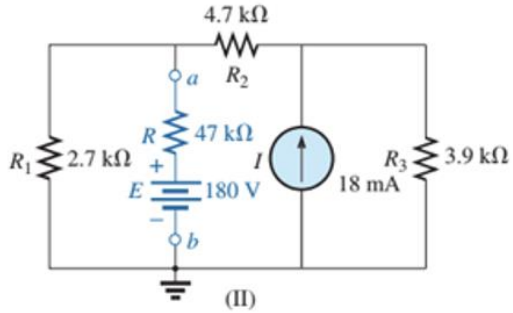


$$V_{6\Omega} = \frac{6\,\Omega(18\text{ V})}{6\,\Omega + 3\,\Omega} = 12\text{ V}$$

$$E_{Th} = 72\text{ V} + 12\text{ V} = \mathbf{84\text{ V}}$$

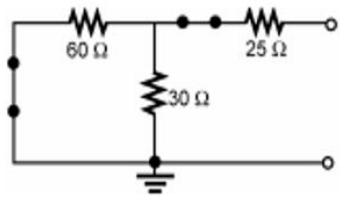
***13.** Find the Thévenin equivalent circuit for the portions of the networks in Fig. 9.131 external to points *a* and *b*.





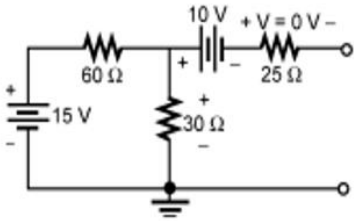
Solution:

(I): R_{Th}:



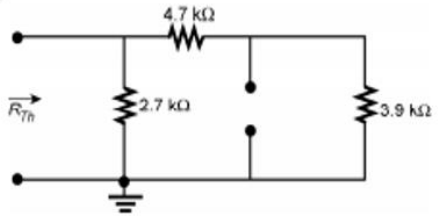
$\leftarrow R_{Th} = 25\ \Omega + 60\ \Omega \parallel 30\ \Omega = \mathbf{45\ \Omega}$

E_{Th}:



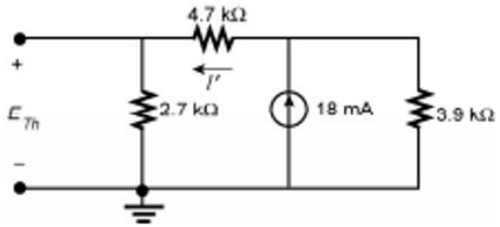
$$E_{Th} = V_{30\Omega} - 10\ \text{V} - 0$$
$$= \frac{30\ \Omega(15\ \text{V})}{30\ \Omega + 60\ \Omega} - 10\ \text{V}$$
$$= 5\ \text{V} - 10\ \text{V} = \mathbf{-5\ V}$$

(II): R_{Th}:



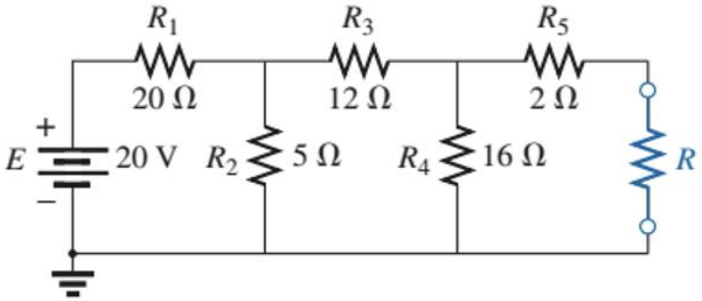
$R_{Th} = 2.7\ \text{k}\Omega \parallel (4.7\ \text{k}\Omega + 3.9\ \text{k}\Omega) = 2.7\ \text{k}\Omega \parallel 8.6\ \text{k}\Omega = \mathbf{2.06\ k\Omega}$

E_{Th}:



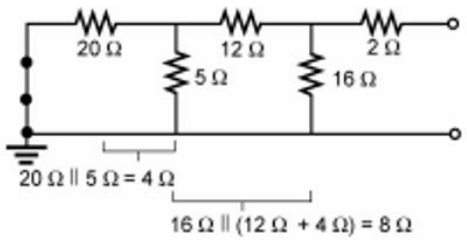
$$I' = \frac{3.9\ \text{k}\Omega(18\ \text{mA})}{3.9\ \text{k}\Omega + 7.4\ \text{k}\Omega} = 6.21\ \text{mA}$$
$$E_{Th} = I'(2.7\ \text{k}\Omega) = (6.21\ \text{mA})(2.7\ \text{k}\Omega) = \mathbf{16.77\ V}$$

*14. Determine the Thévenin equivalent circuit for the network external to the resistor R in both networks in Fig. 9.132.



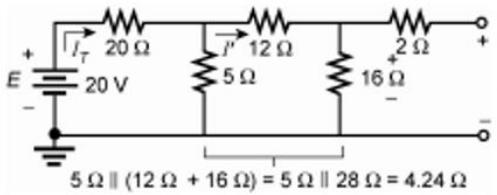
Solution:

(I): R_{Th}:



$\leftarrow R_{Th} = 2\,\Omega + 8\,\Omega = \mathbf{10\,\Omega}$

E_{Th}:



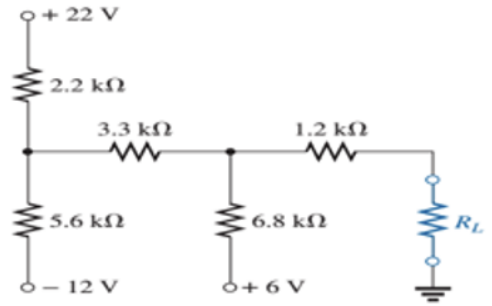
$E_{Th} = V_{16\Omega}$

$$I_T = \frac{20\text{ V}}{20\,\Omega + 4.24\,\Omega} = 825.08\text{ mA}$$

$$I' = \frac{5\,\Omega(I_T)}{5\,\Omega + 28\,\Omega} = \frac{5\,\Omega(825.08\text{ mA})}{33\,\Omega} = 125.01\text{ mA}$$

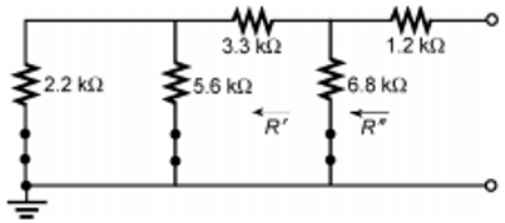
$$E_{Th} = V_{16\Omega} = (I')(16\,\Omega) = (125.01\text{ mA})(16\,\Omega) = \mathbf{2\text{ V}}$$

***15.** For the network in Fig. 9.133, find the Thévenin equivalent circuit for the network external to the load resistor R_L.



Solution:

R_{Th}:



$\leftarrow R_{Th} = 2.2\ \text{k}\Omega \parallel 5.6\ \text{k}\Omega = 1.58\ \text{k}\Omega$

$R' = 1.58\ \text{k}\Omega + 3.3\ \text{k}\Omega$

$= 4.88\ \text{k}\Omega$

$R'' = 4.88\ \text{k}\Omega \parallel 6.8\ \text{k}\Omega = 2.84\ \text{k}\Omega$

$R_{Th} = 1.2\ \text{k}\Omega + R'' = 1.2\ \text{k}\Omega + 2.84\ \text{k}\Omega = \mathbf{4.04\ k\Omega}$

E_{Th}: Source conversions:

$$I_1 = \frac{22\ \text{V}}{2.2\ \text{k}\Omega} = 10\ \text{mA},\ R_s = 2.2\ \text{k}\Omega$$

$$I_2 = \frac{12\ \text{V}}{5.6\ \text{k}\Omega} = 2.14\ \text{mA},\ R_s = 5.6\ \text{k}\Omega$$

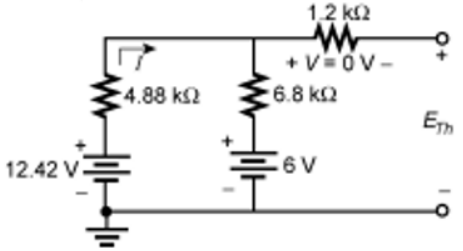
Combining parallel current sources: $I'_T = I_1 - I_2 = 10\ \text{mA} - 2.14\ \text{mA} = 7.86\ \text{mA}$

$2.2\ \text{k}\Omega \parallel 5.6\ \text{k}\Omega = 1.58\ \text{k}\Omega$

Source conversion:

$E = (7.86\ \text{mA})(1.58\ \text{k}\Omega) = 12.42\ \text{V}$

$R' = R_s + 3.3\ \text{k}\Omega = 1.58\ \text{k}\Omega + 3.3\ \text{k}\Omega = 4.88\ \text{k}\Omega$



$$I = \frac{12.42\ \text{V} - 6\ \text{V}}{4.88\ \text{k}\Omega + 6.8\ \text{k}\Omega} = \frac{6.42\ \text{V}}{11.68\ \text{k}\Omega} = 549.66\ \mu\text{A}$$

$V_{6.8\text{k}\Omega} = I(6.8\ \text{k}\Omega) = (549.66\ \mu\text{A})(6.8\ \text{k}\Omega) = 3.74\ \text{V}$

$E_{Th} = 6\ \text{V} + V_{6.8\text{k}\Omega} = 6\ \text{V} + 3.74\ \text{V} = \mathbf{9.74\ V}$

Thank You


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